Introduction to JAVA

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About Java

- ***Java*** is a powerful and versatile programming language for developing software running on mobile devices, desktop computers, and servers
- Developed by James Gosling in 1995
- Popular because of write a program once and run it anywhere
- Java is a **programming language** and a **platform**
- **Platform**: Any hardware or software environment in which a program runs, is known as a platform. Since Java has a runtime environment (JRE) and API, it is called a platform.

Types of Java Applications

- Four types of applications that can be created using Java programming
  - Standalone Application
  - Web Application
  - Enterprise Application
  - Mobile Application

JDK Editions

- Java Standard Edition (Java SE)
- Java Enterprise Edition (Java EE)
- Java Micro Edition (Java ME)

JDK Versions

- History of JDK version
  - https://en.wikipedia.org/wiki/Java_version_history
- **Latest is Java SE 21[Java SE 21]**
- Download from:
  - https://www.oracle.com/java/technologies/downloads/#jdk20-windows

Installing Java

- To check if Java is installed (open command prompt and type following command)

  `java –version`

- Setting Java Path
  - Temporary Path
    - Open the command prompt
    - Copy the path of the JDK/bin directory
    - Write in command prompt: **set path=copied_path**
  - Permanent Path using Environment Variables Settings

Environment Variable

- Go to "System Properties" (Can be found on Control Panel > System and Security > System > Advanced System Settings)
- Click on the "Environment variables" button under the "Advanced" tab
- Then, select the "Path" variable in ***System variables*** and click on the "***Edit***" button
- Click on the "***New***" button and add the path where Java is installed, followed by **\bin**. By default, Java is installed in C:\Program Files\Java\jdk-15\bin
  Then, click "OK", and save the settings
- At last, open Command Prompt (cmd.exe) and type **java -version** to see if Java is running on your machine

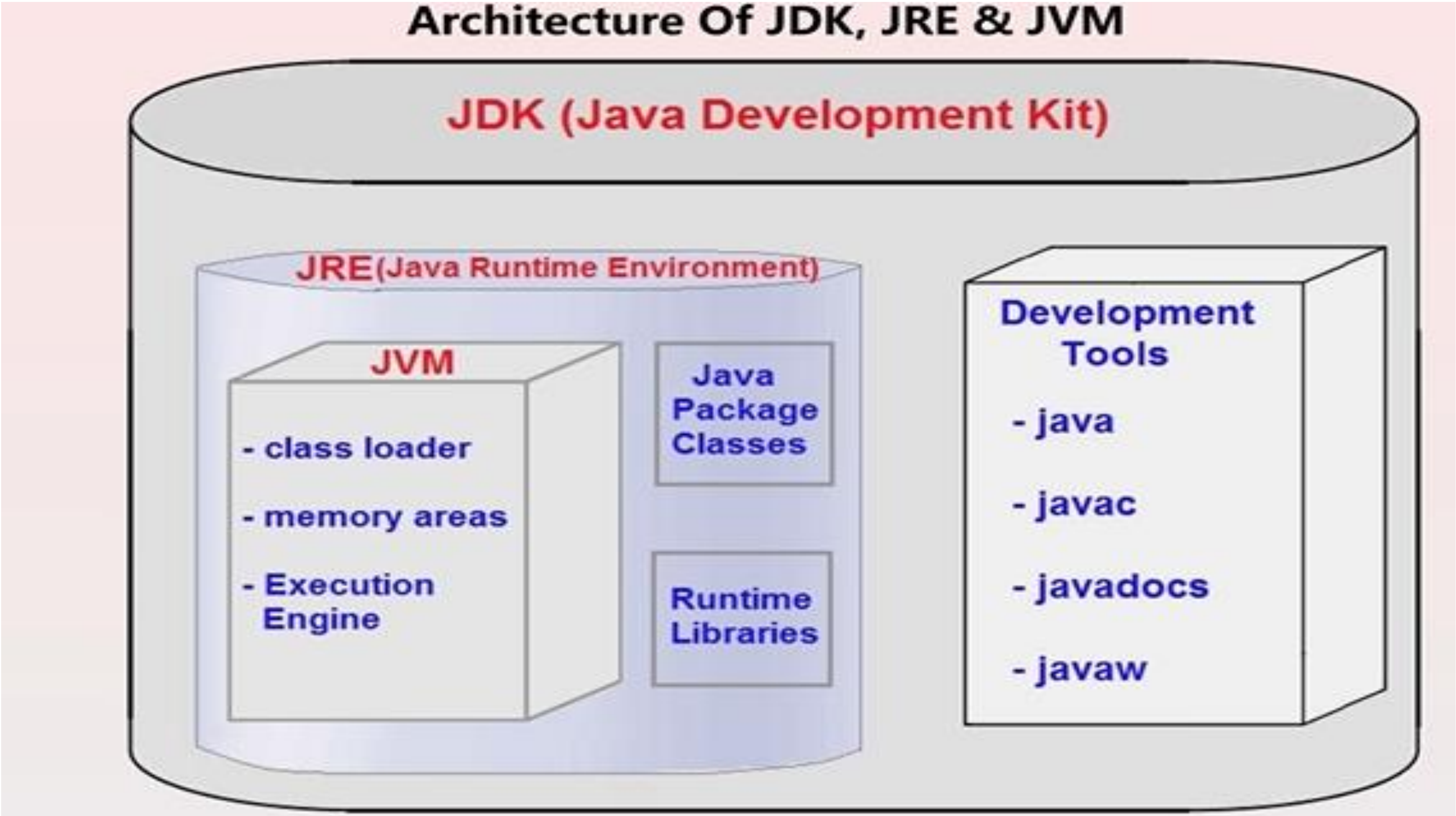
Popular Java IDEs

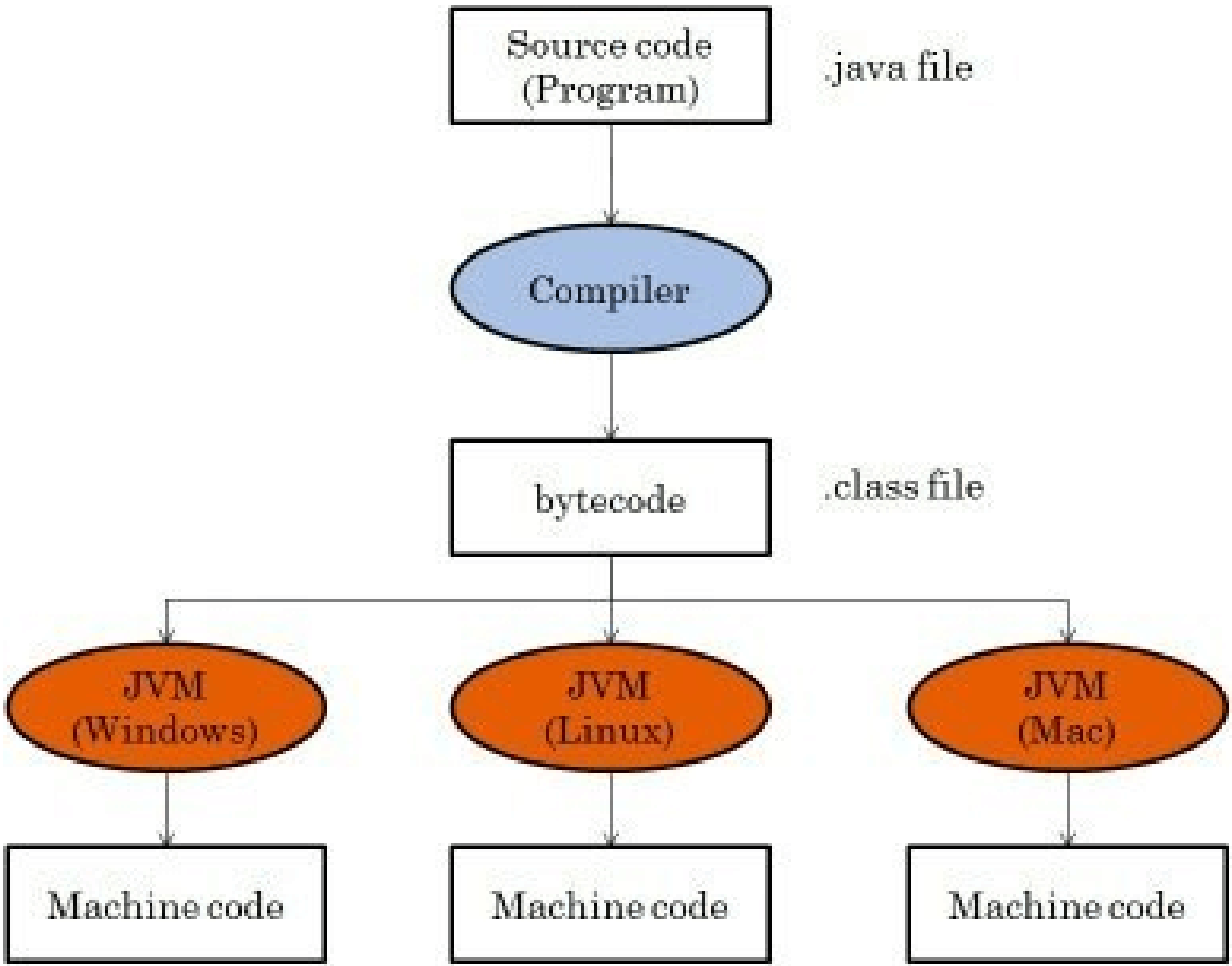
- NetBeans
- Eclipse
- IntelliJ

JDK, JRE, and JVM

- Java Virtual Machine (JVM)
  - Provides a platform-independent way of executing Java source code
  - JVM comes with JIT(Just-in-Time) compiler that converts Java source code into low-level machine language
- Java Runtime Environment
  - Set of software tools which are used for developing Java applications
  - Contains JVM
  - JRE = JVM + rt.jar (lang, util, awt, swing, math packages)
- Java Development Tool kit (JDK)
  - Software development environment which is used to develop Java applications
  - Physically exists

JDK, JRE, and JVM



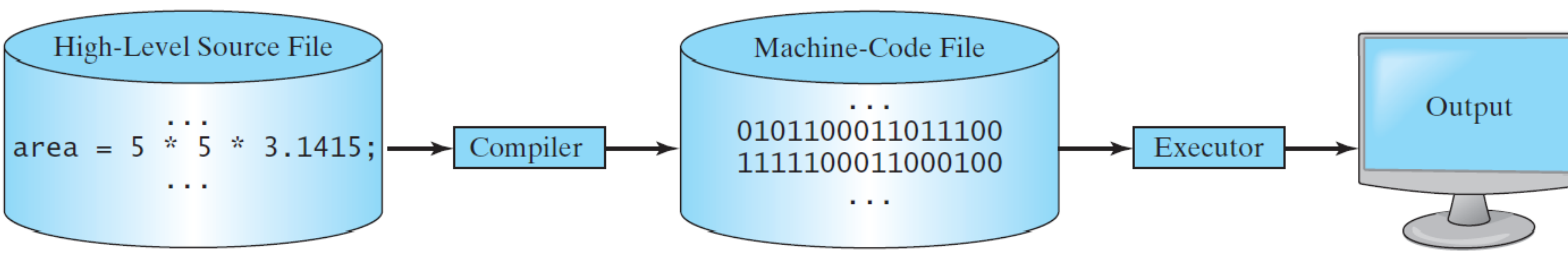

Source code
(Program)
.java file
Compiler
bytecode
.class file
JVM
(Windows)
JVM
(Linux)
JVM
(Mac)
Machine code
Machine code
Machine code

Interpreting/Compiling Source Code

- A program written in a high-level language is called a *source program* or *source code*
- *A source program must be translated into machine code for execution*

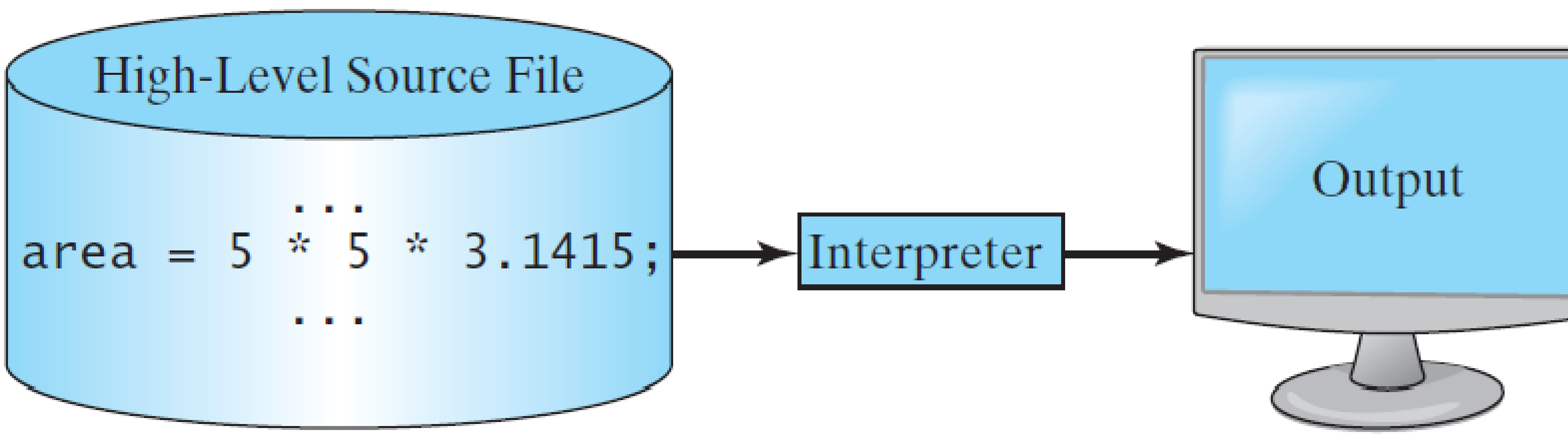
Compiling Source Code

A compiler translates the entire source code into a machine-code file, and the machine-code file is then executed

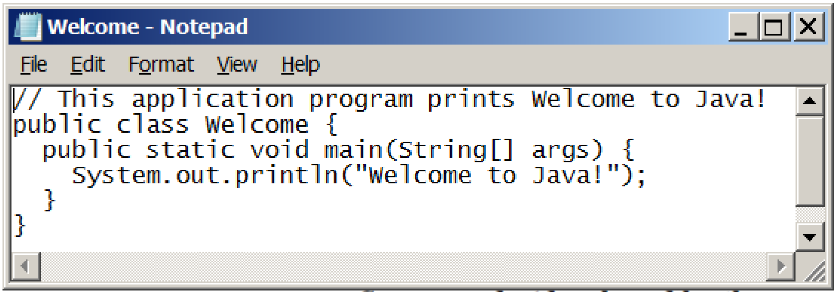


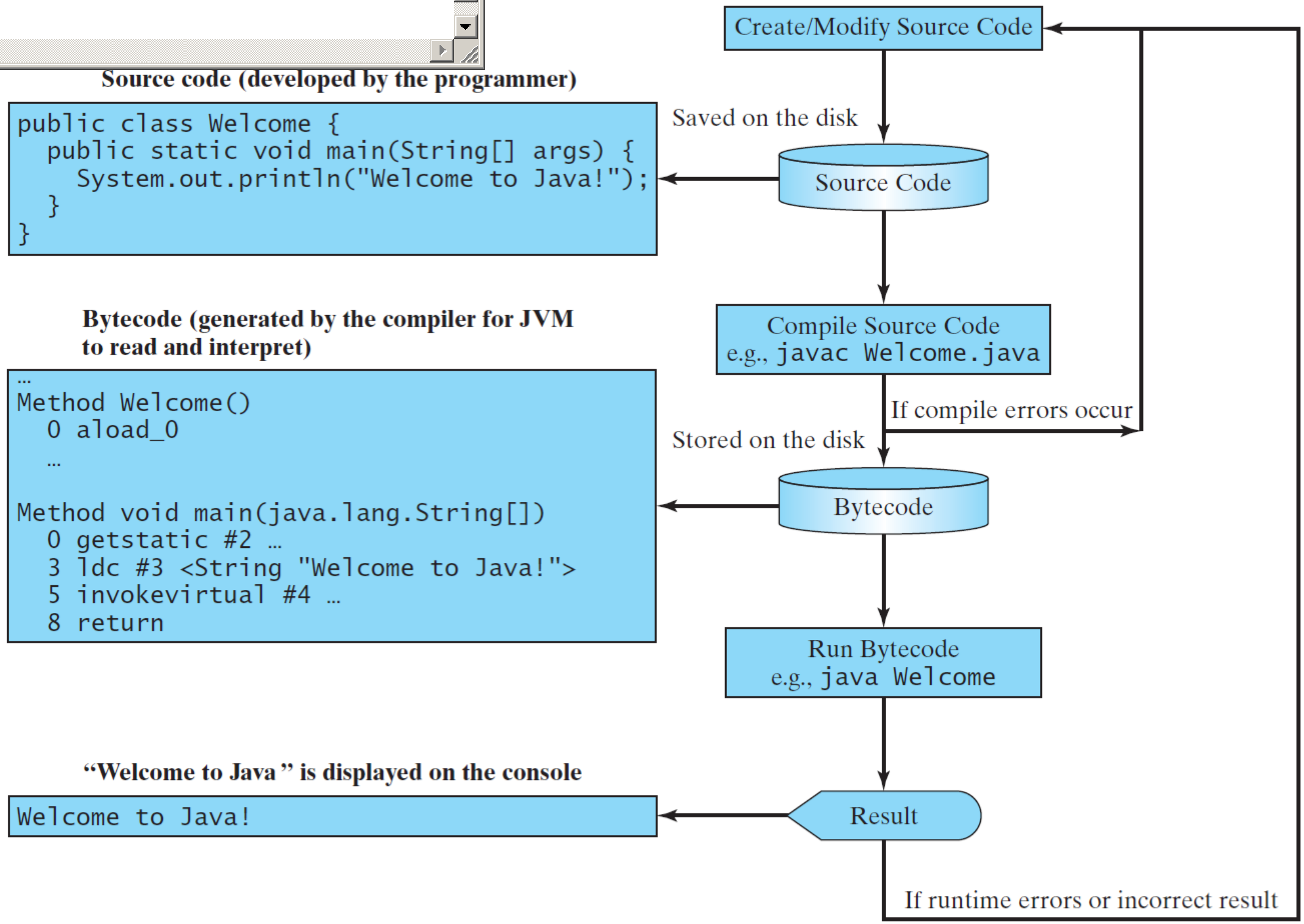
Interpreting Source Code

An interpreter reads one statement from the source code, translates it to the machine code or virtual machine code, and then executes it right away



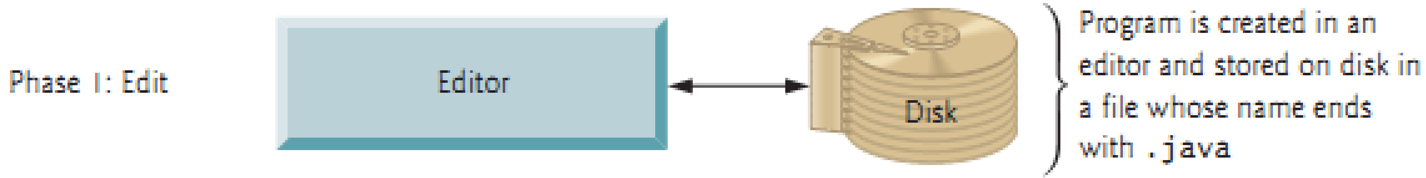
Creating, Compiling, and Running Programs



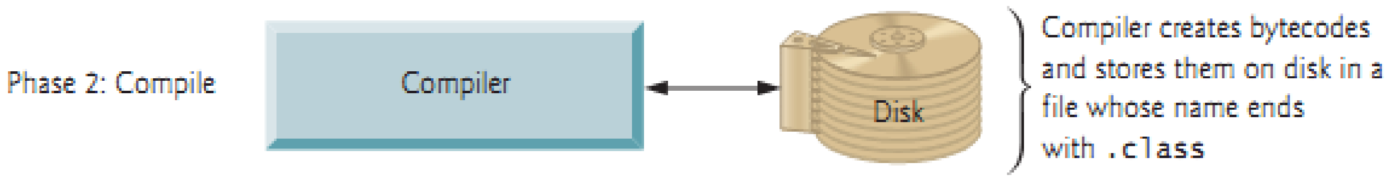


Java Program Phases

- Phase 1: Creating a Program



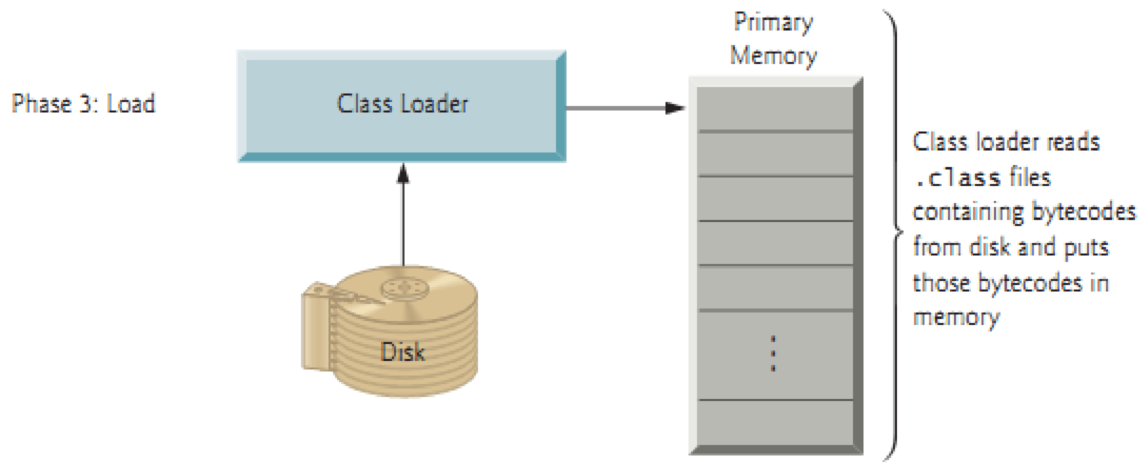
- Phase 2: Compiling a Java Program into Bytecodes



  – Bytecodes are executed by the JVM

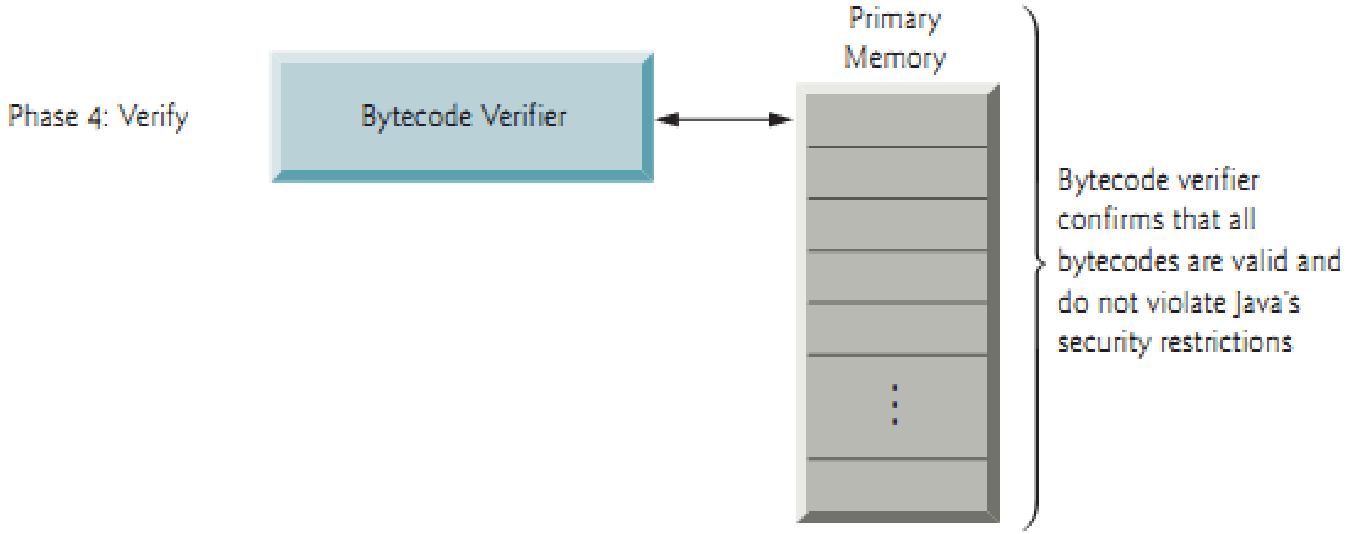
Java Program Phases

- Phase 3: Loading a Program into Memory
  - JVM class loader takes the .class file (along with associated class files provided by Java) and loads in memory



Java Program Phases

- Phase 4: Bytecode Verification
  - Java enforces strong security to make sure that Java programs arriving over the network do not damage your files or your system (as computer viruses and worms might).



Java Program Phases

- Phase 5: Execution
  - Interpreter
    - JVM execute bytecode using interpreter (one bytecode at a time) which slows execution
  - Just In Time(JIT) Compiler
    - JVM analyzes the bytecodes as they're interpreted searching for ***hot spots — parts*** of the bytecodes that execute frequently
    - For these part JIT also known as Java **HotSpot compiler**
      - Translates the bytecodes into the underlying computer's machine language